

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405552
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Itabashi; Nao et al.

Developing cartridge capable of reducing size of image forming apparatus

Abstract

A developing cartridge may include: a casing configured to accommodate developer therein, the casing extending in the first direction; a coupling rotatable about a first axis extending in the first direction, the coupling being positioned at one side of the casing in the first direction; a detection gear rotatable about a second axis extending in the first direction, the detection gear being positioned at another side of the casing in the first direction; and a storage medium including an electric contact surface, the electric contact surface being positioned at the one side of the casing in the first direction.

Inventors:	Itabashi; Nao (Nagoya, JP), Yokoi; Junichi (Nagoya, JP)
Applicant:	BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)
Family ID:	59087833
Assignee:	BROTHER KOGYO KABUSHIKI KAISHA (Nagoya, JP)
Appl. No.:	19/077437
Filed:	March 12, 2025

Prior Publication Data

Document Identifier	Publication Date
US 20250208534 A1	Jun. 26, 2025

Foreign Application Priority Data

JP	2015-254201	Dec. 25, 2015
----	-------------	---------------

Related U.S. Application Data

continuation parent-doc US 18919749 20241018 PENDING child-doc US 19077437
continuation parent-doc US 18633697 20240412 US 12235597 20250225 child-doc US 18919749
continuation parent-doc US 18301293 20230417 US 11988978 20240521 child-doc US 18633697
continuation parent-doc US 17568337 20220104 US 11675289 20230613 child-doc US 18301293
continuation parent-doc US 17077147 20201022 US 11221567 20220111 child-doc US 17568337
continuation parent-doc US 16867612 20200506 US 11067919 20210720 child-doc US 17077147
continuation parent-doc US 16593623 20191004 US 10649369 20200512 child-doc US 16867612
continuation parent-doc US 16361519 20190322 US 10444667 20191015 child-doc US 16593623
continuation parent-doc US 16031011 20180710 US 10254681 20190409 child-doc US 16361519
continuation parent-doc US 15280614 20160929 US 10025225 20180717 child-doc US 16031011

Publication Classification

Int. Cl.: **G03G15/08** (20060101); **G03G21/16** (20060101); G03G21/18 (20060101)

U.S. Cl.:

CPC **G03G15/0867** (20130101); **G03G15/0863** (20130101); **G03G21/1647** (20130101);
G03G21/1896 (20130101)

Field of Classification Search

CPC: G03G (15/0863); G03G (21/1652); G03G (15/80); G03G (21/1867); G03G (21/1871);
G03G (21/1875); G03G (21/1878); G03G (21/1885); G03G (2221/166)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6289189	12/2000	Numagami	399/117	G03G 21/186
7623255	12/2008	Simpson	713/168	G06F 21/34
12235597	12/2024	Itabashi	N/A	G03G 15/0867
2006/0193646	12/2005	Suzuki	399/12	G03G 15/0855
2015/0063833	12/2014	Shinagawa	399/27	G03G 15/556
2017/0184998	12/2016	Itabashi	N/A	G03G 21/1652

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2014063080	12/2013	JP	N/A

Primary Examiner: Aydin; Sevan A

Attorney, Agent or Firm: Merchant & Gould P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 18/919,749, filed Oct. 18, 2024, which is a continuation of U.S.

patent application Ser. No. 18/633,697, filed Apr. 12, 2023, now U.S. Pat. No. 12,235,597, which is a continuation of U.S. patent application Ser. No. 18/301,293, filed Apr. 17, 2023, now U.S. Pat. No. 11,988,978, which is a continuation of U.S. patent application Ser. No. 17/568,337, filed Jan. 4, 2022, now U.S. Pat. No. 11,675,289, which is a continuation of U.S. patent application Ser. No. 17/077,147, filed Oct. 22, 2020, now U.S. Pat. No. 11,221,567, which is a continuation of U.S. patent application Ser. No. 16/867,612, filed May 6, 2020, now U.S. Pat. No. 11,067,919, which is a continuation of U.S. patent application Ser. No. 16/593,623, filed Oct. 4, 2019, now U.S. Pat. No. 10,649,369, which is a continuation of U.S. patent application Ser. No. 16/361,519, filed Mar. 22, 2019, now U.S. Pat. No. 10,446,667, which is a continuation of U.S. patent application Ser. No. 16/031,011, filed Jul. 10, 2018, now U.S. Pat. No. 10,254,681, which is a continuation of U.S. patent application Ser. No. 15/280,614, filed Sep. 29, 2016, now U.S. Pat. No. 10,025,225, which claims priority from Japanese Patent Application No. 2015-254201 filed on Dec. 25, 2015. The contents of all prior applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a developing cartridge.

BACKGROUND

(2) Conventionally, a developing cartridge is capable of being attached to an image forming apparatus such as a laser printer is known. One conventional developing cartridge accommodates developer therein. The developing cartridge includes an electrode for receiving electric power from the image forming apparatus. The electrode can supply a developing roller with the received electric power from the image forming apparatus. The developing cartridge also includes a coupling. The coupling is configured to receive driving force from the image forming apparatus. The one conventional developing cartridge further includes a detection gear. The detection gear is a gear for transmit, to the image forming apparatus, information representing one or more of specifications of the developing cartridge or information representing as to whether the developing cartridge is a new developing cartridge or not. After the developing cartridge is attached to the image forming apparatus, the coupling receives the driving force from the image forming apparatus, and then, the detection gear can rotate.

(3) Another conventional developing cartridge includes a storage medium (e.g., IC chip). The storage medium can store information representing, for example, a number of printing pages. The another developing cartridge includes a cartridge-side electrical contact portion provided at a contact portion of the detection gear.

SUMMARY

(4) In the other conventional developing cartridge, the coupling, the detection gear and the IC chip are positioned at a first side surface of a casing of the developing cartridge, and the first side surface is positioned at one side of the casing. In this case, the first side surface becomes larger because a lot of components (e.g., the coupling, the detection gear and the IC chip) have to be positioned at the first side surface. That causes the developing cartridge to have a large size. Alternatively, the image forming apparatus also includes a driving unit configured to transmit the driving force to the coupling, an actuator configured to detect the detection gear, and an electrical contact portion configured to read information from the IC chip at an inner surface of the image forming apparatus and the inner surface faces the first side surface of the developing cartridge. Therefore, configurations of the inner surface facing the first side surface may be complex structures and that may cause the image forming apparatus to have a large size.

(5) It is an object of the present disclosure to arrange the coupling, the detection gear and an electric contact surface of the storage medium (e.g., IC chip) in an appropriate manner at the developing cartridge in order that a size of the image forming apparatus becomes smaller. Other

objects, features, and advantages will be apparent to persons of ordinary skill in the art from the following detailed description of the disclosure and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a developing cartridge;
- (2) FIG. 2 is a perspective view of the developing cartridge;
- (3) FIG. 3 is a perspective view of the developing cartridge;
- (4) FIG. 4 is an exploded perspective view of an IC (Integrated Circuit) chip assembly;
- (5) FIG. 5 is a perspective view of the developing cartridge;
- (6) FIG. 6 is a perspective view of the developing cartridge;
- (7) FIG. 7 is a perspective view of a developing cartridge and a drum cartridge according to a modification;
- (8) FIG. 8 is a view for illustrating attachment of the drum cartridge to an image forming apparatus in a state where the developing cartridge is attached to the drum cartridge according to the modification.

DETAILED DESCRIPTION OF EMBODIMENTS

(9) A preferred embodiment of the present invention will be described with reference to drawings. In the following embodiment, an extending direction of a rotation axis of a developing roller will be referred to as a “first direction”. A direction perpendicular to the first direction will be referred to as a “second direction”. Specifically, the second direction is defined by a virtual line connecting an agitator shaft **21** of an agitator and a roller shaft **32** of a developing roller **30**.

(10) Overall Structure of Developing Cartridge

(11) FIGS. 1 to 5 are perspective views of a developing cartridge **1**. FIG. 6 is a perspective view of positions of a plurality of gears relative to the developing cartridge **1**. The developing cartridge **1** is used for an electro-photographic type image forming apparatus (for example, a laser printer or a LED printer), and is a unit for supplying developer (toner, for example) to a photosensitive drum. As shown in FIG. 1, the developing cartridge **1** is attached to a drawer unit **90** of the image forming apparatus.

(12) When the developing cartridge **1** is replaced, the drawer unit **90** is drawn out from a front surface of the image forming apparatus. The drawer unit **90** includes four cartridge holding portions **91**, and the developing cartridge **1** is attached to four cartridge holding portions **91**, respectively. Each of four cartridge holding portions **91** includes a photosensitive drum.

(13) In the present embodiment, four developing cartridges **1** are attached to one drawer unit **90**. Each of the four developing cartridges **1** is configured to accommodate developer therein, and the color of the developer is different colors (cyan, magenta, yellow, and black, for example) among the four developing cartridges respectively. However, the number of developing cartridges **1** that can be attached to the drawer unit **90** may be 1 to 3 or be greater than or equal to 5.

(14) As shown in FIGS. 2 to 5, each developing cartridge **1** according to the present embodiment includes a casing **10**, an agitator **20**, a developing roller **30**, a first gear portion **40**, a second gear portion **50**, and an IC (Integrated Circuit) chip assembly **60**.

(15) The casing **10** is a case configured to accommodate therein developer (toner, for example) for electro-photographic printing. The casing **10** includes a first outer surface **11** and a second outer surface **12**. The first outer surface **11** is an outer surface being positioned at one side of the casing **10** in the first direction. The second outer surface **12** is an outer surface being positioned at another side of the casing **10** in the first direction. The first outer surface **11** and the second outer surface **12** are separated from each other in the first direction. The first gear portion **40** and the IC chip assembly **60** are positioned at the first outer surface **11**. The second gear portion **50** is positioned at

the second outer surface **12**. The casing **10** extends in the first direction between the first outer surface **11** and the second outer surface **12**. The developing chamber **13** for accommodating the developer is provided in the casing **10**. The first direction may be an extending direction of the casing **10**. The first direction may be an extending direction of the developing roller **30**.

(16) The agitator **20** extends in the first direction. The agitator **20** includes an agitator shaft **21** and an agitation blade **22**. The agitator shaft **21** extends along the rotation axis **81** extending in the first direction. The agitator shaft **21** has a columnar shape extending in the first direction. The agitator shaft **21** is an example of a first shaft. The agitation blade **22** expands outward from the agitator shaft **21** in a radial direction. The agitation blade **22** is positioned inside a developing chamber **13** of the casing **10**.

(17) As shown in FIG. **4**, the agitator shaft **21** includes a first end portion **211** and a second end portion (not shown in the figures). The first end portion **211** is positioned at one end portion of the agitator shaft **21** in the first direction. The first end portion **211** penetrates through the first outer surface **11** of the casing **10**. A first agitator gear **44** described later is mounted to the first end portion **211**. More specifically, the first agitator gear **44** is mounted to the first end portion **211** so as to be incapable of rotating relative to the first end portion **211**. The first agitator **44** is positioned at the first outer surface **11**.

(18) The second end portion is positioned at another end portion of the agitator shaft **21** in the first direction. The second end portion penetrates through the second outer surface **12** of the casing **10**. A second agitator gear **51** described later is mounted to the second end portion. More specifically, the second agitator gear **51** is mounted to the second end portion so as to be incapable of rotating relative to the second end portion. The second agitator **51** is positioned at the second outer surface **12**.

(19) Accordingly, the agitator shaft **21** and the agitation blade **22** are rotatable with the first agitator gear **44** and the second agitator gear **51**. The developer which is accommodated in the developing chamber **13** is agitated by rotation of the agitation blade **22**. Instead of the agitation blade **22**, the agitator may include an agitation film.

(20) The developing roller **30** is rotatable about a rotation axis **82** extending in the first direction. The developing roller **30** according to the present embodiment includes a roller body **31** and a roller shaft **32**. The roller body **31** is a cylinder-shaped member extending in the first direction. The roller body **31** is made of an elastic rubber, for example. The roller shaft **32** is a cylindrical member penetrating through the roller body **31** in the first direction. The roller shaft **32** is made of metal or conductive resin. The roller body **31** is fixed to the roller shaft **32** so as to be incapable of rotating relative to the roller shaft **32**. When the roller shaft **32** rotates, the roller body **31** rotates together with the roller shaft **32**.

(21) The roller shaft **32** may not penetrate through the roller body **31** in the first direction. For example, each of a pair of roller shafts **32** may extend from each end of the roller body **31** in the first direction.

(22) One end portion of the roller shaft **32** in the first direction is mounted to a developing gear **42** described later so as to be incapable of rotating relative to the developing gear **42**. Accordingly, the roller shaft **32** rotates with rotation of the developing gear **42** and the roller body **31** also rotates with the roller shaft **32**, when the developing gear **42** rotates.

(23) The second direction may be defined by a direction parallel to a virtual line connecting the agitator shaft **21** and the roller shaft **32** at the same position in the first direction. The roller shaft **32** is positioned at one side of the agitator shaft **21** in the second direction. The agitator shaft **21** is positioned at another side of the roller shaft **32** in the second direction. The second direction is perpendicular to the first direction.

(24) The casing **10** has an opening **14**. The opening **14** communicates between the developing chamber **13** and an exterior of the developing chamber **13**. The opening **14** is positioned at one end portion of the casing in the second direction. The developing roller **30** is positioned at the opening

14. That is, the developing roller **30** is positioned closer to one side of the casing **10** than to the center of the casing **10** in the second direction.

(25) When the developing cartridge **1** receives a driving force, the developer is supplied from the developing chamber **13** in the casing **10** onto an outer peripheral surface of the developing roller **30** via a supply roller (omitted in the figure). At this time, the developer is tribocharged between the supply roller and the developing roller **30**. On the other hand, bias voltage is applied to the roller shaft **32** of the developing roller **30**. Accordingly, static electricity between the roller shaft **32** and the developer moves the developer toward the outer peripheral surface of the roller body **31**.

(26) The developing cartridge **1** further includes a layer thickness regulation blade which is omitted in the figure. The layer thickness regulation blade regulates a thin layer of the developer supplied onto the outer peripheral surface of the roller body **31** so that the thickness of the developer becomes constant. Then, the developer on the outer peripheral surface of the roller body **31** is supplied to the photosensitive drum of the drawer unit **90**. At this time, the developer moves from the roller body **31** to the photosensitive drum on the basis of an electrostatic latent image formed on the outer peripheral surface of the photosensitive drum. Accordingly, the electrostatic latent image is visualized on the outer peripheral surface of the photosensitive drum.

(27) The first gear portion **40** is positioned at one end portion in the first direction of the casing **10**. That is, the first gear portion **40** is positioned at the first outer surface **11**. FIG. **4** is a perspective view of the developing cartridge **1** in a state in which the first gear portion **40** is disassembled. As shown in FIG. **4**, the first gear portion **40** includes a coupling **41**, a developing gear **42**, an idle gear **43**, a first agitator gear **44**, and a first cover **45**. As shown in FIG. **4**, the coupling **41**, the developing gear **42**, the idle gear **43** and the first agitator gear are positioned at the first outer surface **11**. A plurality of gear teeth of each gear are not illustrated in FIG. **4**.

(28) The coupling **41** is a gear for initially receiving the driving force applied from the image forming apparatus. The coupling **41** is rotatable about a rotation axis **83** extending in the first direction. The rotational axis **83** which is a rotational center of the coupling **41** is one example of a first axis. The coupling **41** includes a coupling portion **411** and a coupling gear **412**. The coupling portion **411** and the coupling gear **412** are integral with each other and made of a resin, for example.

(29) The coupling portion **411** has a coupling hole **413** recessed in the first direction. The coupling hole **413** is an example of a recessed portion configured to receive driving force from the image forming apparatus. Instead of the coupling hole **413**, the coupling portion **411** may have a concave portion which is configured to receive driving force from the image forming apparatus. The coupling gear **412** includes a plurality of gear teeth. The gear teeth are provided on the entire outer peripheral surface of the coupling gear **412** at equal intervals.

(30) When the drawer unit **90** to which the developing cartridge **1** is attached is accommodated in the image forming apparatus, a drive shaft of the image forming apparatus is inserted into the coupling hole **413** of the coupling portion **411**. With this configuration, the drive shaft and the coupling portion **411** are connected so as to be incapable of rotating relative to each other. Accordingly, the coupling portion **411** rotates when the drive shaft rotates, and the coupling gear **412** rotates together with the coupling portion **411**.

(31) The developing gear **42** is a gear for rotating the developing roller **30**. The developing gear **42** is rotatable about a rotation axis extending in the first direction. The developing gear **42** includes a plurality of gear teeth. The gear teeth are provided on the entire outer peripheral surface of the developing gear **42** at equal intervals. At least a portion of the plurality of gear teeth of the coupling gear **412** meshes with at least a portion of the plurality of gear teeth of the developing gear **42**. Further, the developing gear **42** is mounted to the end portion of the roller shaft **32** in the first direction so as to be incapable of rotating relative to the roller shaft **32**. With this construction, when the coupling gear **412** rotates, the developing gear **42** rotates with the coupling gear **412** and the developing roller **30** also rotates with the developing gear **42**.

(32) The idle gear **43** is a gear for transmitting rotational driving force of the coupling gear **412** to the first agitator gear **44**. The idle gear **43** is an example of a first idle gear. The idle gear **43** is rotatable about a rotation axis **84** extending in the first direction. As shown in FIG. **6**, the idle gear **43** includes a large diameter gear portion **431** and a small diameter gear portion **432**. The large diameter gear portion **431** and the small diameter gear portion **432** are arranged in the first direction. The small diameter gear portion **432** is positioned at another side of the large diameter gear portion **431** in the first direction. That is, the small diameter gear portion **432** is positioned between the large diameter gear portion **431** and the first outer surface **11** of the casing **10**. In other words, the large diameter gear portion **431** is farther away from the first outer surface **11** than the small diameter gear portion **432** is. A diameter of the small diameter gear portion **432** is smaller than a diameter of the large diameter gear portion **431**. In other words, a diameter of an addendum circle of the small diameter gear portion **432** is smaller than a diameter of an addendum circle of the large diameter gear portion **431**. The large diameter gear portion **431** and the small diameter gear portion **432** are integral with each other and are made of a resin.

(33) The large diameter gear portion **431** includes a plurality of gear teeth, and the plurality of gear teeth are provided on the entire outer peripheral surface of the large diameter gear portion **431** at equal intervals. The small diameter gear portion **432** includes a plurality of gear teeth, and the plurality of gear teeth are provided on the entire outer peripheral surface of the small diameter gear portion **432** at equal intervals. The number of gear teeth of the small diameter gear portion **432** is less than the number of gear teeth of the large diameter gear portion **431**. At least a portion of the plurality of gear teeth of the coupling gear **412** meshes with at least a portion of the plurality of gear teeth of the large diameter gear portion **431**. Further, at least a portion of the plurality of gear teeth of the small diameter gear portion **432** meshes with at least a portion of the plurality of gear teeth of the first agitator gear **44**. When the coupling gear **412** rotates, the large diameter gear portion **431** rotates together with the coupling gear **412** and the small diameter gear portion **432** rotates together with the large diameter gear portion **431**. Also, the first agitator gear **44** rotates with the rotation of the small diameter gear portion **432**.

(34) The first agitator gear **44** is a gear for rotating the agitator **20** in the developing chamber **13**. The first agitator gear **44** is rotatable about a rotation axis **81** extending in the first direction. The first agitator gear **44** includes a plurality of gear teeth, and the plurality of gear teeth are provided on the entire outer peripheral surface of the first agitator gear **44** at equal intervals. As described above, the at least a portion of the plurality of gear teeth of the small diameter gear portion **432** meshes with the at least a portion of the plurality of gear teeth of the first agitator gear **44**. Further, the first agitator gear **44** is mounted to one end portion of the agitator shaft **21** in the first direction so as to be incapable of rotating relative to the agitator shaft **21**. With the configuration, when the rotational driving force is transmitted from the coupling **41** to the first agitator gear **44** via the idle gear **43**, the first agitator gear **44** rotates and the agitator **20** rotates together with the first agitator gear **44**. That is, the agitator **20** including the agitator shaft **21** rotates together with the coupling **41**.

(35) In the developing cartridge **1** of this embodiment, the idle gear **43** is positioned between the coupling gear **412** and the first agitator gear **44**, however, the idle gear **43** may be omitted. That is, the coupling gear **412** may directly mesh with the first agitator gear **44**.

(36) The first cover **45** is positioned at one side of the casing in the first direction. More specifically, the first cover is positioned at the first outer surface. The first cover **45** is fixed to the first outer surface **11** of the casing **10** by screws, for example. The coupling gear **412**, the developing gear **42**, the idle gear **43**, and the first agitator gear **44** are accommodated in a space between the first outer surface **11** and the first cover **45**. The coupling hole **413** of the coupling portion **411** is exposed to an outside of the first cover **45**. The first cover **45** according to the present embodiment also serves as a holder cover for holding the holder **62** of the IC chip assembly **60** described later. A structure of the first cover **45** as the holder cover will be described later in detail.

(37) The second gear portion **50** is positioned at the other side of the casing **10** in the first direction. In other words, the second gear portion **50** is positioned at the second outer surface **12**. FIG. 5 is a perspective view of the developing cartridge **1** in which the second gear portion **50** is exploded. As illustrated in FIG. 5, the second gear portion **50** includes a second agitator gear **51**, a detection gear **52**, an electrically conductive member **53**, and a second cover **54**. Note that, in FIG. 5, gear teeth are not illustrated in the second agitator gear **51** and the detection gear **52**.

(38) The second agitator gear **51** is for transmitting rotational driving force of the agitator shaft **21** to the detection gear **52**. The second agitator gear **51** is rotatable about a rotation axis **81** extending in the first direction. The second agitator gear **51** includes a plurality of gear teeth, and the plurality of gear teeth are provided on the entire outer peripheral surface of the second agitator gear **51** at equal intervals. At least a portion of the plurality of gear teeth of the second agitator gear **51** meshes with at least a portion of a plurality of gear teeth of the detection gear **52**. The second agitator gear **51** is mounted to the first end portion of the agitator shaft **21** in the first direction so as to be incapable of rotating relative to the agitator shaft **21**. With this configuration, the second agitator gear **51** rotates with rotation of the agitator shaft **21**.

(39) The detection gear **52** is a gear for providing information on the developing cartridge **1** for the image forming apparatus. The information on the developing cartridge **1** includes, for example, information as to whether the developing cartridge **1** is a new (unused) cartridge or a used cartridge. The information on the developing cartridge **1** also includes, for example, a product specification of the developing cartridge **1**. The product specification of the developing cartridge **1** includes, for example, the number of sheets that can be printed with the developer accommodated in the developing cartridge **1** (i.e. sheet-yield number).

(40) The detection gear **52** is rotatable about a rotation axis **85** extending in the first direction. The rotational axis **85** which is a rotational center of the detection gear **52** is an example of a second axis. The detection gear **52** includes a plurality of gear teeth **521**. The gear teeth **521** are provided on a portion of an outer peripheral surface of the detection gear **52**. That is, the detection gear **52** is a tooth-less gear, that is, the plurality of gear teeth **521** are provided on one portion of an outer peripheral surface of the detection gear **52**. The other portion of the outer peripheral surface of the detection gear **52** does not include a gear tooth.

(41) When the developing cartridge **1** is in an unused state, at least a portion of the plurality of gear teeth of the detection gear **52** can mesh with at least a portion of the plurality of gear teeth of the second agitator gear **51**. In this case, the detection gear **52** rotates together with the agitator **20** including agitator shaft **21**. For this reason, the detection gear **52** rotates based on the driving force transmitted via the coupling **41** receives driving force, the coupling **41**, the idle gear **43**, the first agitator gear **44**, the agitator **20** and the second agitator gear **51**. That is the detection gear **52** is rotatable with the coupling **41**.

(42) When the image forming apparatus starts to operate, the developing cartridge **1** is attached to the drawer unit **90** and the drawer unit is inserted into the inside of the image forming apparatus and accommodated in the inside of the image forming apparatus. When the drawer unit **90** to which an unused developing cartridge **1** is attached is attached in the image forming apparatus, the coupling **41** receives driving force and then, the detection gear **52** can rotate by meshing with the second agitator gear **51**. When the detection gear **52** rotates at a predetermined angle, the detection gear **52** is disengaged from the second agitator gear **51**, rotation of the detection gear **52** is stopped.

(43) When the developing cartridge **1** is in the unused state, the detection gear is in a first position representing that the developing cartridge **1** is in the unused state. When the detection gear **52** is in the first position, at least a portion of the plurality of gear teeth of the detection gear **52** can mesh with at least a portion of the plurality of gear teeth of the second agitator gear **51**. When the developing cartridge **1** starts to work in the image forming apparatus, the detection gear **52** rotates from the first position and a second position. Therefore, the detection gear **52** is in the second position representing that the developing cartridge **1** is an used state, after the developing cartridge

1 starts to work. When the detection gear **52** is in the second position, the detection gear **52** does not mesh with the second agitator gear **51**. Thus, the detection gear **52** can change between the unused state and the used state and then, the detection gear **52** cannot rotate.

(44) Further, the detection gear **52** may be configured of a movable gear that can move in the first direction. The movable gear may not be limited to a partially toothless gear. In other words, the movable gear includes a plurality of gear teeth, and the plurality of gear teeth are provided on an outer peripheral surface of the movable gear along the circumference of the movable gear. In this case, the movable gear moves in the first direction in accordance with rotation of the movable gear, thereby the movable gear is disengaged from the second agitator gear **51**. The movable gear may be moved in the first direction away from the second outer surface **12** or toward the second outer surface **12**.

(45) When the drawer unit **90** to which a used developing cartridge **1** is attached is attached in the image forming apparatus, the detection gear **52** cannot rotate because the detection gear **52** is disengaged from the second agitator gear **51**.

(46) A gear may be provided between the second agitator gear **51** and the detection gear **52**. For example, the second gear portion **50** may further include a second idle gear meshing with both the second agitator gear **51** and the detection gear **52**. The second idle gear is positioned at the second outer surface **12**. In this case, rotational driving force of the second agitator gear **51** may be transmitted to the detection gear **52** via the second idle gear.

(47) The electrically conductive member **53** is electrically conductive. The electrically conductive member **53** is an example of an electrode. The electrically conductive member **53** is formed of a material such as electrically conductive metal or electrically conductive resin. The electrically conductive member **53** is positioned at the second outer surface **12** of the casing **10**. The electrically conductive member **53** includes a gear shaft **531** protruding in the first direction. The gear shaft **531** is positioned at the second outer surface **12**. The gear shaft **531** extends in the first direction from the second outer surface **12** along the rotational axis **85**. The rotational axis **85** is an example of a second axis. The gear shaft **531** is an example of a second shaft. The detection gear **52** rotates about the gear shaft **531** in a state where the detection gear **52** is supported by the gear shaft **531**. The electrically conductive member **53** further includes a bearing portion **532**. The bearing portion **532** is in contact with the roller shaft **32** of the developing roller **30**.

(48) A portion of the electrically conductive member **53** may be in contact with the roller shaft **32**. Alternatively, the roller shaft **32** may be in contact with the electrically conductive member **53** in a state where the roller shaft **32** is inserted into the electrically conductive member **53**.

(49) The drawer unit **90** includes an electrically conductive lever (not illustrated) that is in contact with the gear shaft **531** in a state where the developing cartridge **1** is attached to the drawer unit **90**. Instead of the drawer unit **90**, the image forming apparatus may include the electrically conductive lever. When the lever contacts the gear shaft **531**, electrical connection between the lever, and the electrically conductive member **53** is established and electrical connection between the electrically conductive member **53** and the roller shaft **32** is also established. When the image forming apparatus is in operation, electric power is supplied to the roller shaft **32** through the lever, and the roller shaft **32** can keep a prescribed bias voltage. That is, the electrically conductive member **53** including the gear shaft **531** has a function of the electrode supplying the roller shaft **32** with the bias voltage (electric power).

(50) The second cover **54** is positioned at the other side of the casing **20** in the first direction. More specifically, the second cover **54** is positioned at the second outer surface **12**. The second cover **54** is fixed to the second outer surface **12** of the casing **10** by a screw, for example. At least a portion of one or more of the second agitator gear **51** and the detection gear **52**, and the electrically conductive member **53** are accommodated in a space between the second outer surface **12** and the second cover **54**. Therefore, the second cover **54** covers at least a portion of the detection gear **52**. The second cover **54** has an opening **541**. A portion of the detection gear **52** and a portion of the

gear shaft **531** are exposed to the outside through the opening **541**. The electrically conductive lever of the drawer unit **90** contacts the detection gear **52** and the gear shaft **531** through the opening **541**.

(51) As illustrated in FIG. 5, the detection gear **52** includes a detecting protrusion **522**. The detection gear **52** covers a portion of an outer peripheral surface of the gear shaft **531**. The protrusion **522** is positioned at another side the plurality of gear teeth **521** in the first direction. The detecting protrusion **522** protrudes in the first direction. The detecting protrusion **522** has a circular arc shape extending along a portion of an addendum circle of the detection gear about the rotation axis of the detection gear **52**. Note that the detecting protrusion **522** covers a portion of an outer peripheral surface of the gear shaft **531**. The detecting protrusion **522** is rotatable with the detection gear **52**.

(52) When the developing cartridge in the unused state is attached to the image forming apparatus, a portion of the gear shaft **531** is exposed to the outside through the opening **541**. That is, the lever of the drawer unit **90** is in contact with the gear shaft **531**. When the image forming apparatus is in operation and the coupling **41** receives driving force, the detection gear **52** rotates. Then, the detecting protrusion **522** pass through between the lever and the gear shaft **531** according to the rotation of the detection gear **52**. The lever is not in contact with the gear shaft **531**, when the detecting protrusion **522** is positioned between the lever and the gear shaft **531**. After the detection gear **42** further rotates, the detecting protrusion **522** pass through between the lever and the gear shaft **531** and the lever is in contact with the gear shaft **531**. When the detection gear **52** rotates at a predetermined angle, the detection gear **52** is disengaged from the second agitator gear **51**, rotation of the detection gear **52** is stopped. Therefore, the contact state between the lever and the gear shaft **531** is maintained.

(53) Hence, when the detection gear **52** rotates after a new developing cartridge **1** is attached in the drawer unit **90**, the contact state between the lever and the gear shaft **531** changes according to the shape of the detection gear **52**. More specifically, the contact state between the lever and the gear shaft **531** changes according to the shape of the detecting protrusion **522** because the detecting protrusion **522** pass through between the lever and the gear shaft according to the rotation of the detection gear **52**. Alternatively, the contact state between the lever and the gear shaft **531** changes according to the number of the detecting protrusions **522** which are provided with the detection gear **52** because one or more of detecting protrusions **522** pass through between the lever and the gear shaft according to the rotation of the detection gear **52**. The image forming apparatus recognizes the change in the contact state between the lever and the gear shaft **531** to identify whether the attached developing cartridge **1** is new or used and/or the product specification of the mounted developing cartridge **1**. That is, the detection gear **52** has a shape representing information regarding a specification of the developing cartridge. For example the specification of the developing cartridge may represent a color of the developer accommodated in the developing cartridge. The detection gear **52** may have a shape representing other information representing a color of the developer accommodated in the developing cartridge.

(54) However, the method for detecting the information on the developing cartridge **1** using the detection gear **52** is not limited to detection of electrical conduction. For example, movement of the lever may be optically detected. Further, the detecting protrusion **522** may be formed to have different circumferential position and length from those in the present embodiment. Further, the detection gear **52** may have a plurality of detecting protrusions **522**. The shape of the detection gear **52** may vary according to the product specification of the developing cartridge **1** such as the number of printable sheets. More specifically, the number of the detecting protrusions **522** may be differentiated among a plural type of the developing cartridges, and the product specification regarding each of the developing cartridges may be identified based on the number of the detecting protrusions **522**. When each of the plural type of the developing cartridges includes the number of the detecting protrusions **522**, circumferential intervals between the plurality of detecting

protrusions **522** may be differentiated among the plural type of the developing cartridges. In the above-described case, a circumferential length of each detecting protrusion **522** and/or a radial length of each detecting protrusion **522** may be differentiated based on the product specification regarding each of the developing cartridges. In this way, variations in the number of the detecting protrusions **522** and/or circumferential positions of each of the detecting protrusion **522** enables the image forming apparatus to identify the product specification regarding each of the developing cartridges.

(55) The detection gear **52** may be configured of plural components. For example, the detecting protrusion **522** and the detection gear **52** may be different components. Further, the detection gear **52** may include a detection gear body and a supplemental member that shifts its position relative to the detection gear body in accordance with rotation of the detection gear body. In this case, the supplemental member changes between a first position in which the supplemental member is in contact with the lever and a second position in which the supplemental member is not in contact with the lever in accordance with shifting the position of the supplemental member relative to the detection gear body. As a result, the supplemental member may change the position of the lever.

(56) Further, the detection gear **52** may include a cam, and the cam may contact the detecting protrusion **522**. In this case, the cam rotates together with rotation of the detection gear **52**, and the rotating cam contacts the detecting protrusion **522**. This causes the detecting protrusion **522** to move relative to the detection gear **52**. The detecting protrusion **522** may be rotatably attached to a shaft provided at the second outer surface **12** or the second cover **54**. Alternatively, the detecting protrusion **522** may have a shaft, and the shaft of the detecting protrusion **522** may be inserted into a hole formed in the second outer surface **12** or the second cover **54** so that the detecting protrusion **522** is rotatably supported by the second outer surface **12** or the second cover **54**.

(57) Further, in the present embodiment, the gear shaft **531** extends in the first direction from the second outer surface **12**. However, the gear shaft **531** does not need to be in direct contact with the second outer surface **12**. For example, the casing **10** may have a through-hole penetrating the second outer surface **12** and a cap attached or fitted with the through-hole, and a gear shaft may extend from the cap in the first direction. In this case, the cap includes the gear shaft protruding in the first direction toward the detection gear **52**, and the detection gear **52** rotates about the gear shaft **531** in a state where the detection gear is supported by the gear shaft **531**.

(58) 2. IC Chip Assembly

(59) The IC chip assembly **60** is positioned at the one side of the casing in the first direction. The IC chip assembly **60** is positioned at the first outer surface **11** of the casing **10**. FIG. **6** is an exploded perspective view of the IC chip assembly **60**. FIG. **7** is a cross-sectional view of the IC chip assembly **60** taken along a plane perpendicular to the first direction. As shown in FIGS. **2** through **7**, the IC chip assembly **60** includes an IC (Integrated Circuit) chip **61** as a storage medium and a holder **62** for holding the IC chip **61**. The holder **62** is held to the first cover **45** at one end of the casing **10** in the first direction. The IC chip **61** stores various information on the developing cartridge **1**.

(60) As shown in FIG. **5**, the IC chip **61** includes an electric contact surface **611**. The electric contact surface **611** is made of electrically conductive metal. The electric contact surface **611** is positioned at one side of the casing **10** in the first direction. The electric contact surface is positioned at the first outer surface.

(61) The drawer unit **90** includes an electric connector. The electric connector is made of metal, for example. The electric connector of the drawer unit **90** contacts the electric contact surface **611** when the developing cartridge **1** is attached to the drawer unit **90**. At this time, the image forming apparatus can perform at least one of reading information from the IC chip **61** and writing information in the IC chip **61**.

(62) In this developing cartridge **1**, both the IC chip **61** and the electric contact surface **611** of the IC chip **61** are positioned at the one side of the casing in the first direction.

(63) At least a portion of the holder **62** is covered by the first cover **45**. The holder **62** includes a boss **621a**, a boss **621b**, and a boss **621c**. Each of the boss **621a** and boss **621b** extends in the first direction toward the first cover **45** from a surface of the holder **62** opposite to a surface thereof facing the casing **10**. The boss **621a** and boss **621b** are aligned in the second direction. The boss **621c** extends in the first direction toward the casing **10** from the surface of the holder **62** facing the casing **10**. As shown in FIGS. **2** and **4**, the first cover **45** has a through-hole **451a** and a through-hole **451b**. The through-hole **451a** and through-hole **451b** penetrate the first cover **45** in the first direction, respectively. The through-hole **451a** and through-hole **451b** are aligned in the second direction. On the other hand, the casing **10** includes a recessed portion **15**. The recessed portion **15** is recessed in the first direction on the first outer surface **11** of the casing **10**.

(64) The boss **621a** is inserted into the through-hole **451a**. The boss **621b** is inserted into the through-hole **451b**. The boss **621c** is inserted into the recessed portion **15**. The through-hole **451a** has a dimension (inner dimension) larger than a dimension (outside dimension) of the boss **621a**. The through-hole **451b** has a dimension (inner dimension) larger than a dimension (outside dimension) of the boss **621b**. Further, the recessed portion **15** has a dimension (inner dimension) larger than a dimension (outer dimension) of the boss **621c**. Hence, the holder **62** can move with the bosses **621a**, **621b** and **621c** in direction perpendicular to the first direction relative to the casing **10** and the first cover **45**. The holder **62** moves between the first cover **45** and the first outer surface **11**.

(65) Alternatively, the holder **62** may include a single boss, or three or more bosses. Likewise, the first cover **45** may have a single through-hole, or three or more through-holes. The bosses **621a**, **621b** and **621c** may have a circular columnar shape or a rectangular columnar shape, respectively.

(66) Or, instead of the through-holes **451a** and **451b**, the first cover **45** may include one or more of recesses to have the bosses **621a** and/or **621b** inserted therinto.

(67) A projected area of the developing cartridge **1** in the first direction should be smaller in order to down size of the image forming apparatus. That is the first outer surface **11** should be smaller in order to down size of the image forming apparatus. That is the second outer surface **12** also should be smaller in order to down size of the image forming apparatus. On the other hand, it is difficult to arrange the coupling **41**, the electric contact surface **611** and the detection gear **52** at one side of the casing **10** in the first direction in a state where at least a portion of the coupling **41**, the electric contact surface **611** and the detection gear **52** are overlapping in the first direction, because the coupling **41**, each of the electric contact surface **611** and the detection gear is a component for interacting with the image forming apparatus.

(68) As shown in the FIG. **6**, in the developing cartridge **1**, the detection gear **52** is positioned at the other side of the casing **10** in the first direction, and the coupling **41** and the electric contact surface **611** of the IC chip **61** are also positioned at the one side of the casing **10** in the first direction. Therefore, the first outer surface **11** and the second outer surface **12** become smaller because the coupling **41** and the electric contact surface **611** are positioned at the first outer surface **11** and the detection gear **52** is positioned at the second outer surface **12** which is different from the first outer surface **11**.

(69) In this embodiment, the coupling **41** is positioned at one side of the agitator shaft **21** in the second direction, and the coupling **41** is positioned at the one side of the casing in the first direction. The detection gear **52** is positioned at the one side of the agitator shaft **21** in the second direction and the detection gear is positioned at the other side of the casing in the first direction. Therefore, the coupling **41** and the detection gear are positioned at the same side of the agitator shaft **21** in the second direction. For this reason, a length of the developing cartridge in the second direction can be shortened. Accordingly, the developing cartridge **1** can be downsized in the second direction. The image forming apparatus can also be downsized in the second direction.

(70) Specifically, in this embodiment, the detection gear **52** and the electrically conductive member **53** should be the same outer surface (either the first outer surface **11** or the second outer surface **12**)

of the casing **10**, because the detection gear **52** is supported by the electrically conductive member **53**. On the other hand, the electrically conductive member **53** (an example of the electrode) receives high electrical voltage (electrical power). For this reason, IC chip **61** is affected by, for example, high-frequency noise, if a distance between the electrically conductive member **53** and the electric contact surface **611** of the IC chip **61** is too short or a distance between the electrically conductive member **53** and the IC chip **61** is too short.

(71) In this embodiment, in the developing cartridge **1**, the detection gear **52** and the electrically conductive member **53** are positioned at the second outer surface **12** which is positioned at opposite side of the first outer surface **11** in the first direction, and one or more of the electric surface **611** and the IC chip **61** is positioned at the first outer surface **11**. Accordingly, an electrical interference between the between the electrically conductive member **53** and the electric contact surface **611** can be reduced. Alternatively, an electrical interference between the between the electrically conductive member **53** and the IC chip **61** can be reduced. For example, the high-frequency noise on the IC chip **61** can be reduced based on charging the high electrical voltage (electrical power) to the electrically conductive member **53**.

(72) In this embodiment, a whole of the IC chip **61** including the electric contact surface **611** is positioned at the first outer surface **11** which is opposite side of the second outer surface **12** in the first direction, the detection gear **52** and the electrically conductive member **53** are positioned at the second outer surface **12**. At least the electric contact surface **611** may be positioned at the first outer surface **11** and the IC chip **61** is positioned at a different position from the first outer surface (e.g., another surface of the casing **10**). An electrical interference between the between the electrically conductive member **53** and the electric contact surface **611** can be reduced, if at least the electric contact surface **611** may be positioned at the first outer surface **11** and the IC chip **61** is positioned at a different position from the first outer surface (e.g., another surface of the casing **10**). Alternatively, an electrical interference between the between the electrically conductive member **53** and the IC chip **61** can be reduced. For example, the high-frequency noise on the IC chip **61** can be reduced based on charging the high electrical voltage (electrical power) to the electrically conductive member **53**.

(73) 3. Modifications

(74) While the description has been made in detail with reference to the specific embodiment thereof, it would be apparent to those skilled in the art that various changes and modifications may be made therein. In the following description, differences between the above embodiment and the modifications are mainly explained.

(75) FIG. 7 is a perspective view showing a developing cartridge **1A** and a drum cartridge **70A** of a modification. The developing cartridge **1A** shown in FIG. 7 includes a casing **10A**, a developing roller **30A**, a coupling **41A** a detection gear **52A**, and an IC chip **61A**. In the embodiment shown in FIG. 7, the developing cartridge **1A** is attached to the drum cartridge **70A** instead of the drawer unit. The drum cartridge **70A** includes one developing cartridge holding portion **71A** holding the developing cartridge **1A**. The developing cartridge holding portion **71A** includes a photosensitive drum **72A**. When the developing cartridge **1A** is attached to the drum cartridge **70A**, the developing roller **30A** of the developing cartridge **1A** is in contact with the photosensitive drum **72A**.

(76) FIG. 8 is a view showing how to attach the drum cartridge **70A** to an image forming apparatus **100A** in a state where the developing cartridge **1A** is attached to the drum cartridge **70A**. As shown in FIG. 8, the drum cartridge **70A** is attached to a drum cartridge holding portion **101A** provided in the image forming apparatus **100A** in a state where the developing cartridge **1A** is attached to the drum cartridge **70A**.

(77) In the above manner, a similar structure to that of the developing cartridge **1** according to the above embodiment can be applied to the developing cartridge **1A** to be attached to the drum cartridge **70A**. Specifically, as shown in FIG. 7, in the developing cartridge **1A**, the coupling **41A** and an electric contact surface **611A** of the IC chip **61A** are positioned at the one side of the casing

10A in the first direction, and the detection gear **52A** is also positioned at the other side of the casing **10A** in the first direction. Therefore, the one outer surface and the other outer surface separated from the one outer surface in the first direction become smaller because the coupling **41A** and the electric contact surface **611A** are positioned at the one outer surface and the detection gear **52A** is positioned at the other outer surface which is different from the one outer surface. Accordingly, the developing cartridge **1A** can be downsized. The image forming apparatus **100A** can also be downsized.

(78) In this embodiment, the gear shaft **531** (an example of the second shaft) extends in the first direction from the second outer surface **12**. The gear shaft **531** may not directly contact with the second outer surface **12**. For example, the casing **10** may have a through-hole penetrating the second outer surface **12** and a cap being attached to the through-hole. The gear shaft may extend from the cap in the first direction.

(79) In this case the cap may include the gear shaft protruding in the first direction toward the detection gear **52**. The detection gear **52** may be rotatable about the gear shaft in a state where the detection gear **53** is supported by the gear shaft.

(80) According to the above-described embodiments, the plural gears provided within each of the first gear portion and the second gear portion are engaged with one another through meshing engagement of the gear teeth. However, the plural gears provided within each of the first gear portion and the second gear portion may be engaged with one another through a frictional force. For example, instead of the plural gear teeth, frictional members, such as rubber members, may be provided to the outer circumferences of two gears that engage with each other.

(81) The developing cartridge **1** in this embodiment is attached to the drawer unit of the image forming apparatus. The developing cartridge may be attached to another image forming apparatus which does not include the drawer unit.

(82) Shapes of the details in the developing cartridge may differ from those shown in the drawings attached to this application. The respective components employed in the above-described embodiment and modifications can be selectively combined together within an appropriate range so that no inconsistency will arise.

Claims

1. A developing cartridge comprising: a developing roller rotatable about a first axis extending in a first direction, the developing roller including a roller body extending in the first direction; a casing configured to accommodate developer therein, the casing having a first side and a second side being spaced apart from the first side in the first direction, the casing having a third side and a fourth side being spaced apart from the third side in a second direction, the first side having an outer surface, the developing roller being positioned at the third side of the casing, the casing having a top surface facing upward and a bottom surface facing downward; a coupling rotatable about a second axis extending in the first direction, the coupling being positioned at the outer surface, the second axis being spaced apart from the first axis in the second direction, the coupling including a coupling gear; and a storage medium including an electric contact surface, the electric contact surface being positioned at the outer surface, the electric contact surface facing upward, the electric contact surface being farther from the developing roller than the coupling gear is from the developing roller in the second direction, the electric contact surface being spaced apart from the coupling in the second direction, wherein the electric contact surface is farther from a distal end of the roller body than the outer surface is from the distal end of the roller body in the first direction.
2. The developing cartridge according to claim 1, wherein the developing cartridge does not comprise a photosensitive drum.
3. The developing cartridge according to claim 1, wherein the coupling is spaced apart from the developing roller in the second direction.

4. The developing cartridge according to claim 1, further comprising: an agitator rotatable about a third axis extending in the first direction according to rotation of the coupling, the agitator including an agitator shaft extending in the first direction, the agitator shaft being farther from the developing roller than the second axis is from the developing roller in the second direction.
 5. The developing cartridge according to claim 4, further comprising: an agitator gear rotatable with the agitator shaft, the agitator gear being positioned at the outer surface, the agitator gear being farther from the developing roller than the second axis is from the developing roller in the second direction.
 6. The developing cartridge according to claim 1, wherein the developing roller includes a roller shaft extending in the first direction, the roller shaft being rotatable with the roller body, and wherein the electric contact surface is spaced apart from a distal end of the roller shaft in the second direction.
 7. The developing cartridge according to claim 4, wherein the second direction is defined by connecting a roller shaft of the developing roller and the agitator shaft.
 8. The developing cartridge according to claim 1, further comprising: an idle gear rotatable about an idle axis extending in the first direction, the idle gear being positioned at the outer surface.
 9. The developing cartridge according to claim 8, wherein the idle gear meshes with the coupling gear.
 10. The developing cartridge according to claim 9, wherein the idle gear and the coupling are positioned between the developing roller and the electric contact surface in the second direction.
 11. The developing cartridge according to claim 1, further comprising: a holder holding the electric contact surface, the holder being positioned at the outer surface, the holder being movable relative to the casing.
 12. The developing cartridge according to claim 1, further comprising: a detection protrusion movable according to the rotation of the coupling, the detection protrusion being positioned at the second side of the casing.
 13. The developing cartridge according to claim 1, further comprising: an electrode being positioned at the second side of the casing, the electrode being electrically connected to a component of the developing cartridge.
 14. The developing cartridge according to claim 13, wherein the electrode is electrically connected to a roller shaft of the developing roller.
 15. The developing cartridge according to claim 13, wherein a portion of the electrode is spaced apart from the developing roller in the second direction.
 16. The developing cartridge according to claim 1, wherein the second direction is perpendicular to the first direction.
 17. The developing cartridge according to claim 1, wherein the storage medium is positioned at the outer surface.
 18. The developing cartridge according to claim 1, wherein the storage medium stores information regarding the developing cartridge.
 19. The developing cartridge according to claim 1, wherein the top surface is spaced apart from the bottom surface in a third direction.
 20. The developing cartridge according to claim 19, wherein the third direction is perpendicular to the first direction.
-